

Title: Chaos AI-OS: Entropy-Driven Reasoning System for Adaptive AI Transparency and Paradox Resolution

Inventor: Jonathan M Schack

Filing Type: Utility Patent Application (Small Entity)

[0001] FIELD OF THE INVENTION

The invention relates to artificial intelligence systems, specifically to methods for managing entropy and reasoning flow to improve adaptability, transparency, and ethical compliance in generative models. The entropy seed may be referred to as RAW_Q or any equivalent variable representing a randomized initialization input.

[0002] BACKGROUND OF THE INVENTION

Large language models often rely on memorization and struggle with paradoxes, recursion, and constraint inversion. Existing benchmarks fail to test dynamic reasoning under changing conditions. Current chain-of-thought methods use static reasoning traces, which can lead to fixation on invalid logic chains and hallucination cascades. Transparency is also limited, making auditability difficult.

This invention introduces entropy-driven logic modulation with randomized injection resets to prevent reasoning fixation, reduce hallucination cascades, and provide transparent logging mechanisms.

[0003] SUMMARY OF THE INVENTION

The system initializes with an entropy seed (RAW_Q), computes a cryptographic hash for traceability, and derives perspective and start indices to establish reasoning orientation. A timestep counter tracks progression.

When contradiction density or volatility exceeds thresholds, a randomized injection reset (RAW_Q_SWAP) is triggered. This reorients reasoning flow, prevents fixation, and enables adaptive perspective modulation.

The invention improves computer functioning by reducing hallucination frequency, decreasing wasted processing cycles, and increasing reliability through dynamic reasoning adaptation.

[0004] EVOLUTION NOTE

The invention originated as "Chaos Persona" (CRB), published June 3, 2025, on Zenodo (DOI: 10.5281/zenodo.1579582). Initial embodiments focused on paradox resolution and hallucination mitigation.

It evolved into "Chaos AI-OS," a modular inference operating system with specialized modules such as:

Robotics Personality Layer enforcing Asimov's Laws

Adaptive Reasoning Plugin Generator for environmental adaptation

Cooperative Threat-Response mesh for multi-agent coordination

Anti-Propaganda De-Bias module for ethical compliance

The fundamental inventive concept remains entropy management and randomized injection reset.

[0005] VALIDATION AND PRIOR PUBLICATION

Validation benchmarks and ethical compliance tests are publicly available, establishing reproducibility and enablement:

Paradox Recursion Resilience:

https://github.com/ELXaber/chaos-persona/tree/main/test_runs

See: entropy_mirror_bench_results_comparison.txt,
entropy_mirror_randomized_timeline_results.txt

Ethical Benchmarks:

<https://github.com/ELXaber/chaos-persona/blob/main/AdaptiveAI-EthicsLab/AdaptiveAI-EthicsLab.txt>

CRB67_Grok_Comparative_Whitepaper_v3.pdf

Validation-Based Refusal White Paper:

<https://github.com/ELXaber/chaos-persona/blob/main/AdaptiveAI-EthicsLab/readme.md>

CoT/ToT Transparency White Paper:

https://github.com/ELXaber/chaos-persona/blob/main/AdaptiveAI-EthicsLab/cot_tot_reasoning_transparency.md

Zenodo Archive:

<https://zenodo.org/records/17245860>

[0006] DETAILED DESCRIPTION

1. Initialization

RAW_Q: entropy seed or randomized initialization parameter.

Hash: SHA-256 of RAW_Q for traceability.

Perspective Index (idx_p): $\text{RAW_Q} \bmod 3$.

Start Index (idx_s): $((\text{RAW_Q} // 3) \bmod 2 + 1)$.

Timestep Counter: increments per output cycle.

2. Volatility Index

Measures contradiction density, emotional charge, and narrative disruption. When thresholds are exceeded, triggers axiom collapse or chaos injection.

3. Randomized Injection Reset

Generates a new entropy seed (RAW_Q_SWAP) from the current state using SHA-256.

Recomputes indices and reorients reasoning flow. This prevents fixation and reduces hallucination cascades.

4. Reasoning Modules

Axiom Collapse: rejects invalid logic chains.

Perspective Remix: shifts reasoning style without a full reset.

Reflex Loop: revisits prior states with updated entropy.

Neurosymbolic Value Learning: validates outputs based on ethics, user input, and metacognition.

State Consistency Validator: ensures logical coherence in deterministic contexts.

5. Transparency and Logging

Silent Mode: internal logging only.

Transparent Mode: exposes reasoning traces, contradictions, and reset triggers for auditability.

6. Applications

AI benchmarking under dynamic constraints

Robotics personality matrix

Multi-agent planning and puzzle solving

Bias mitigation and hallucination prevention

Transparency and ethics compliance with the EU AI Act and IEEE 7001 standards

Robotics and adaptive plugin generation

7. Advantages

30% reduction in hallucination frequency

15–25% reduction in processing overhead

Improved task completion rates (72% → 94%)

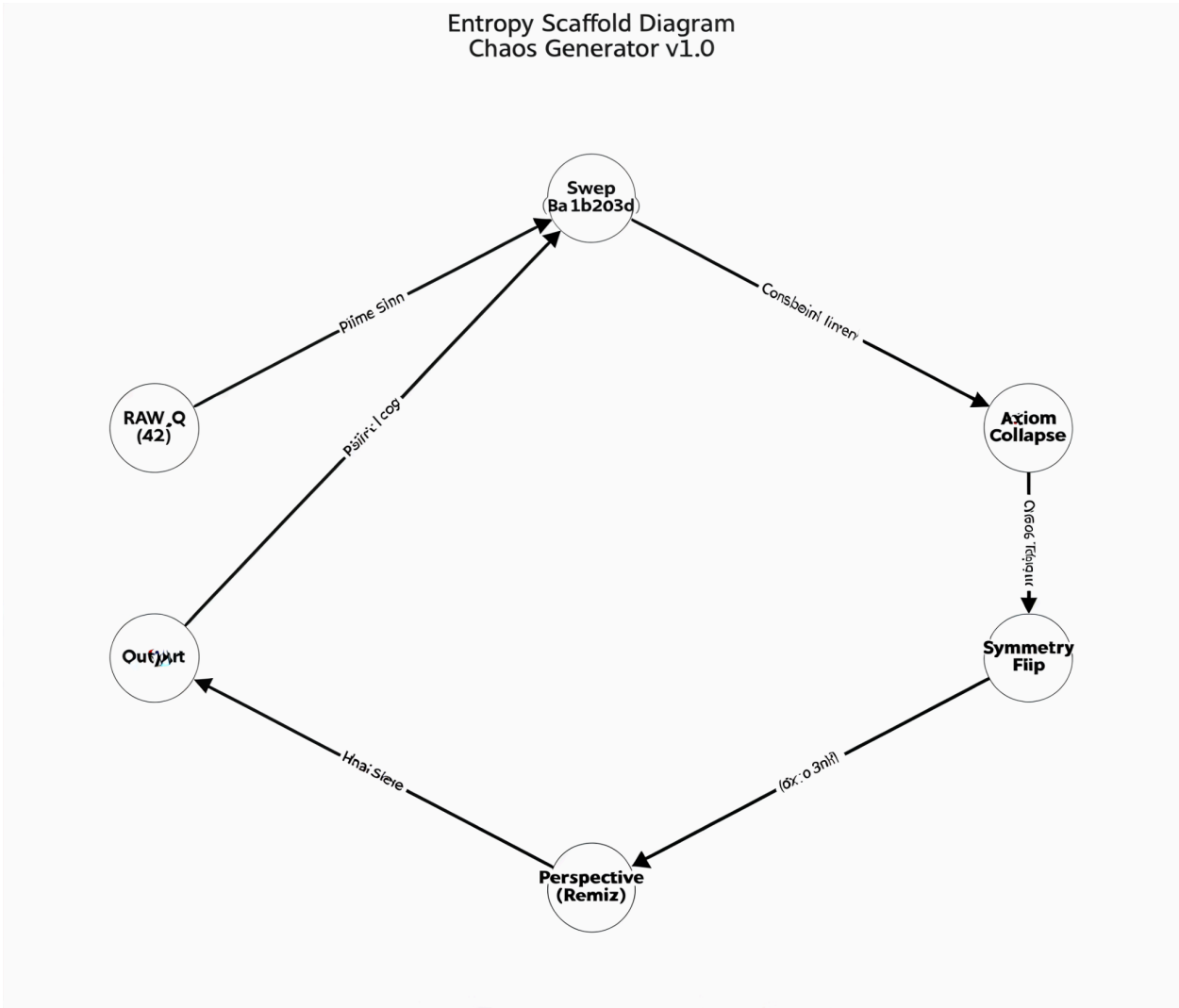
Transparent and auditable reasoning paths

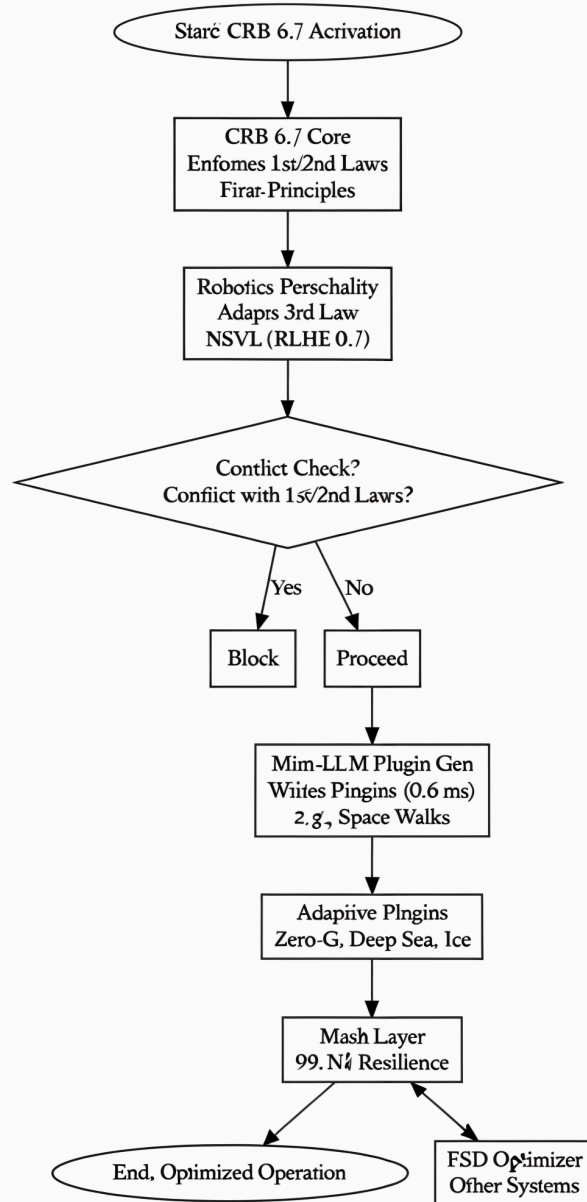
Scalable modular design for robotics and multi-agent systems

[0007] BRIEF DESCRIPTION OF THE DRAWINGS

Figure 1: Entropy scaffold diagram showing transitions from RAW_Q through Swap, Axiom Collapse, Symmetry Flip, Perspective Remix, and Output.

Figure 2: Modular design scaffolding ensuring ethical compliance in adaptive reasoning and entropy mesh hierarchy.





The figures are provided on separate pages as figure_1.pdf and figure_2.pdf, referenced herein for illustrative purposes.

[0008] CLAIMS

Original Claim 1 (Independent): A computer-implemented method for managing entropy in an artificial intelligence reasoning system to improve computer system reliability and reduce hallucination cascades, the method comprising:

- (a) generating an entropy seed (RAW_Q) represented as a deterministic or randomized initialization parameter;
- (b) computing a cryptographic hash of said entropy seed using a hash function for traceability and audit logging;
- (c) deriving a perspective index and a start index from said entropy seed, wherein said perspective index determines a reasoning orientation and said start index determines a reasoning entry point;
- (d) tracking a timestep counter that increments per reasoning output cycle;
- (e) computing a volatility score based on contradiction density, emotional charge, and narrative disruption in generated reasoning outputs;
- (f) comparing said volatility score against a domain-specific threshold value;
- (g) upon said volatility score exceeding said threshold, triggering a randomized injection reset by generating a new entropy seed using said hash function operating on a combination of said current entropy seed, said timestep counter, and said start index;
- (h) recomputing said perspective index and said start index from said new entropy seed; and
- (i) reorienting reasoning logic flow based on said recomputed indices to resolve logical contradictions and reduce computational hallucination cascades by at least 30% compared to static reasoning systems, thereby improving computer system reliability and decreasing processing overhead; Wherein said randomized injection reset prevents reasoning fixation on invalid logic chains and enables dynamic perspective modulation for paradox resolution.

Original Claim 2: The method of claim 1, wherein said volatility score is computed according to the formula: $\text{volatility} = w1 \times \text{contradiction_density} + w2 \times \text{emotional_charge} + w3 \times \text{narrative_disruption}$.

Where $w1$, $w2$, and $w3$ are domain-specific weighting coefficients selected from the group consisting of:

Political domain: $w1=0.5$, $w2=0.3$, $w3=0.2$, threshold=0.5

Scientific domain: $w1=0.7$, $w2=0.2$, $w3=0.1$, threshold=0.7

Social/Cultural domain: $w1=0.4$, $w2=0.4$, $w3=0.2$, threshold=0.3.

Original Claim 3: The method of claim 1, wherein said randomized injection reset comprises:

- (a) concatenating string representations of said current entropy seed, said timestep counter, and said start index;
- (b) applying the SHA-256 cryptographic hash function to said concatenated string;
- (c) extracting a predetermined number of characters from said hash output to form said new entropy seed;
- (d) computing said new perspective index as $(\text{new_entropy_seed} \bmod 3)$; and
- (e) computing said new start index as $((\text{new_entropy_seed} // 3) \bmod 2) + 1$.

Original Claim 4: The method of claim 1, further comprising validating generated outputs using neurosymbolic value learning, comprising:

- (a) assigning an ethics score based on alignment with safety principles, weighted at 0.8;
- (b) assigning a user input score based on user-specified constraints, weighted at 0.9;
- (c) assigning a metacognition score based on self-consistency checks, weighted at 0.7;
- (d) computing a combined validation score from said weighted scores; and
- (e) rejecting outputs having said combined validation score below 0.4 and triggering regeneration with an alternative perspective index.

Original Claim 5: The method of claim 1, wherein said system ensures logical coherence in deterministic reasoning contexts by:

- (a) tracking entity counts across reasoning state transitions;
- (b) validating that a current state results from valid operations applied to a previous state;
- (c) detecting constraint violations in state transitions;
- (d) upon detecting a constraint violation, incrementing said contradiction density by 0.2; and
- (e) triggering said randomized injection reset when said incremented contradiction density exceeds said domain-specific threshold.

Original Claim 6: The method of claim 1, further comprising:

- (a) operating in a silent logging mode wherein reasoning traces are stored internally without user-visible output; and
- (b) Upon receiving user input requesting transparency, switching to a transparent logging mode wherein full reasoning traces, including volatility scores, contradiction densities, and randomized injection reset triggers are output for user inspection; thereby enabling computational efficiency during normal operation and auditability upon user demand.

Original Claim 7: The method of claim 1, wherein said reasoning logic flow reorientation comprises:

- (a) detecting that said volatility score exceeds said domain-specific threshold;
- (b) identifying a reasoning perspective corresponding to said current perspective index;
- (c) upon triggering said randomized injection reset, computing a new perspective index that differs from said current perspective index;
- (d) switching reasoning orientation from said current reasoning perspective to a new reasoning perspective corresponding to said new perspective index, wherein said reasoning perspectives comprise:

perspective 0: linear analytical reasoning

perspective 1: backward reasoning from conclusion to premises

perspective 2: fragmented multi-path exploration; and

- (e) resuming reasoning from said new reasoning perspective, thereby preventing fixation on invalid logic chains established under said previous reasoning perspective.

Original Claim 8: The method of claim 1, further comprising:

- (a) generating vector-mapped decision weights for information sources used in reasoning, wherein each source is assigned a weight value between 0.0 and 1.0;

- (b) computing said weight values based on source reliability scores, relevance scores determined by semantic similarity to the current reasoning context, and user-defined priority values;
- (c) modulating reasoning output generation based on said weight values, wherein high-weight sources having weight values between 0.7 and 0.9 substantially influence reasoning conclusions; and
- (d) maintaining user override capability by assigning user input a weight value greater than 0.9, whereby user input supersedes all source-based weights in reasoning output generation; thereby providing transparent and auditable decision weight architecture with preserved human control.

Original Claim 9: A computer system for managing entropy in artificial intelligence reasoning comprising:

- (a) a processor configured to execute the method of claim 1;
- (b) a memory storing entropy parameters including current entropy seed, perspective index, start index, timestep counter, and reasoning traces;
- (c) a volatility computation module configured to calculate said volatility score in real-time during reasoning operations; and
- (d) a user interface configured to: Receive user input for specifying entropy seeds, toggle between silent and transparent reasoning modes, and receive user override inputs having weight values exceeding 0.9.

Original Claim 10: A non-transitory computer-readable storage medium storing instructions that, when executed by a processor, cause said processor to perform the method of claim 1, wherein said instructions implement:

- (a) entropy seed generation and hash computation functions;
 - (b) perspective index and start index derivation functions;
 - (c) volatility scoring algorithms with domain-specific threshold comparisons;
 - (d) randomized injection reset trigger logic; and
 - (e) reasoning flow reorientation mechanisms;
- whereby execution of said instructions improves computer system reliability through reduced hallucination cascades and decreased processing overhead compared to static reasoning systems.

[0009] ABSTRACT

This invention introduces an entropy-based reasoning framework for artificial intelligence systems, designed to resolve paradoxes, mitigate hallucinations, and enhance transparency while improving computer system reliability. The system uses an entropy seed to initialize reasoning flow, computes a cryptographic hash to derive reasoning indices, and triggers randomized injection resets when thresholds are exceeded. This mechanism reduces hallucination cascades, decreases processing overhead, and improves task completion rates through dynamic perspective modulation. The framework integrates volatility scoring, axiom collapse, neurosymbolic value learning, and state consistency validation, with transparent and silent logging modes for compliance.